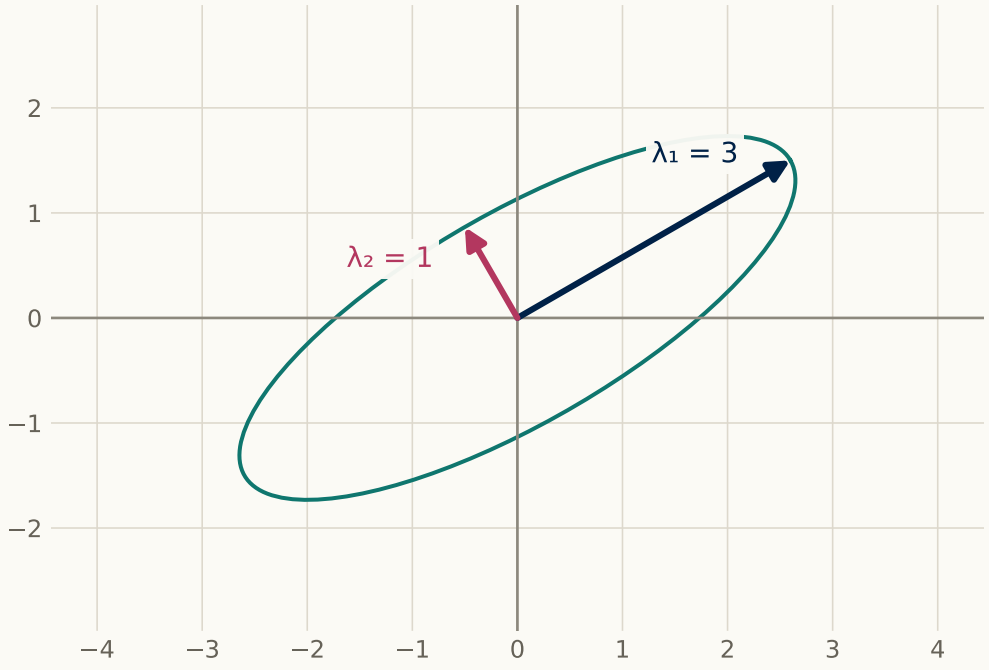
Principal directions disentangle a symmetric rule



Ellipse: image of the unit circle under $S = R\,\mathrm{diag}(3,1)\,R^{T}$.